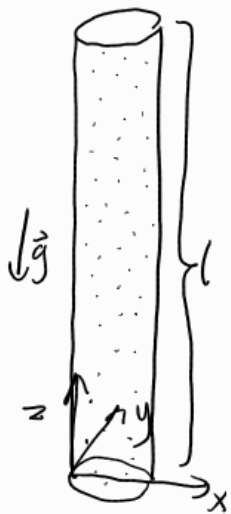


1.1) Enoatomni idealni plin, visok valj, $C = ?$
(pri dani T)



$$C = \frac{d\langle E \rangle}{dT}$$

$$e^{-\beta F} = c \int_{\Gamma} e^{-\beta \sum_{i=1}^N \left(\underbrace{\frac{p_i^2}{2m}}_{\sqrt{\frac{\pi}{2m}} \beta} + \underbrace{mgz_i}_{\frac{1}{-\beta mg}(e^{-\beta mgl} - 1)}} \right)} \underbrace{\prod_{i=1}^N dx_i dy_i dz_i dp_{xi} dp_{yi} dp_{zi}}_A =$$

$$= c \left[\left(\sqrt{\frac{2\pi m}{\beta}} \right)^3 \frac{1 - e^{-\beta mgl}}{\beta mg} A \right]^N = \text{konst.} \cdot \beta^{\frac{5N}{2}} (1 - e^{-\beta mgl})^N$$

$$+\beta F = \ln(\text{konst.}) + \frac{5N}{2} \ln \beta - N \ln(1 - e^{-\beta mgl}) \quad \bigg/ \frac{d}{d\beta}$$

$$\langle E \rangle = \frac{5N}{2\beta} + N \frac{e^{-\beta mgl}(-mgl)}{1 - e^{-\beta mgl}} = \frac{5}{2} \frac{N}{\beta} + \frac{N mgl}{e^{\beta mgl} - 1}$$

Limite:

$\beta mgl \gg 1$: (močen vpliv gravitacije)

$$\langle E \rangle = \frac{5}{2} N k_B T + 0 \rightarrow C_v = \frac{d\langle E \rangle}{dT} = \frac{5 N k_B}{2}$$

$\beta mgl \ll 1$: (tvisoka, zanemarljiva gravitacija)

$$\langle E \rangle = \frac{5}{2} N k_B T - \frac{N mgl}{1 + \beta mgl + \dots - 1} = \frac{5}{2} N k_B T - N k_B T = \frac{3}{2} N k_B T$$

(29) Vdrueno vlakno (veriga molekul keratina)



podatki:

a, ϵ_a N členov

d, ϵ_d

$\langle E \rangle = ?$

$\langle L(T) \rangle = ?$

$\langle L(T, F) \rangle = ?$

$$\langle E \rangle = \frac{d(\beta F)}{d\beta}$$

$$e^{-\beta F} = \sum_{\text{konfiguracije}} e^{-\beta E(N_a)} = \sum_{\text{konf.}} e^{-\beta (\epsilon_a N_a + \epsilon_d (N - N_a))}$$

$$\left. \begin{aligned} E &= \epsilon_a N_a + \epsilon_d N_d \\ N_a + N_d &= N \\ L &= a N_a + d N_d \end{aligned} \right\} \begin{array}{l} \text{fazni prostor za } N=3: \\ \hline \begin{array}{c|c} \text{---} & N_a \\ \hline \text{---} & 3 \\ \hline 1\text{---} & 2 \\ \hline \text{---}1 & 2 \\ \hline \text{---} & 1 \\ \hline 11 & 1 \\ \hline 1-1 & 1 \\ \hline -11 & 1 \\ \hline 111 & 0 \end{array} \end{array}$$

$$Z_N = e^{-\beta F} = \sum_{N_a=0}^N \binom{N}{N_a} e^{-\beta (\epsilon_a N_a + \epsilon_d (N - N_a))} = \sum_{N_a=0}^N \binom{N}{N_a} (e^{-\beta \epsilon_a})^{N_a} (e^{-\beta \epsilon_d})^{N - N_a} =$$

$$= (e^{-\beta \epsilon_a} + e^{-\beta \epsilon_d})^N \rightarrow -\beta F = N \ln(e^{-\beta \epsilon_a} + e^{-\beta \epsilon_d}) \quad \left/ \frac{d}{d\beta} \right.$$

$$\rightarrow Z_N = (Z_1)^N$$

$$\langle E \rangle = -N \frac{-\epsilon_a e^{-\beta \epsilon_a} - \epsilon_d e^{-\beta \epsilon_d}}{e^{-\beta \epsilon_a} + e^{-\beta \epsilon_d}} =$$

$$= N \frac{\epsilon_a e^{-\beta \epsilon_a} + \epsilon_d e^{-\beta \epsilon_d}}{e^{-\beta \epsilon_a} + e^{-\beta \epsilon_d}} = \underbrace{\epsilon_a \langle N_a \rangle + \epsilon_d \langle N_d \rangle}$$

$$\langle L(T) \rangle = a \langle N_a \rangle + d \langle N_d \rangle =$$

$$= N \frac{a e^{-\beta \epsilon_a} + d e^{-\beta \epsilon_d}}{e^{-\beta \epsilon_a} + e^{-\beta \epsilon_d}}$$

$$\epsilon_a \rightarrow \epsilon_a - F a$$

$$\epsilon_d \rightarrow \epsilon_d - F d$$

$$\rightarrow \langle L(T, F) \rangle = N \frac{a e^{-\beta (\epsilon_a - F a)} + d e^{-\beta (\epsilon_d - F d)}}{e^{-\beta (\epsilon_a - F a)} + e^{-\beta (\epsilon_d - F d)}}$$